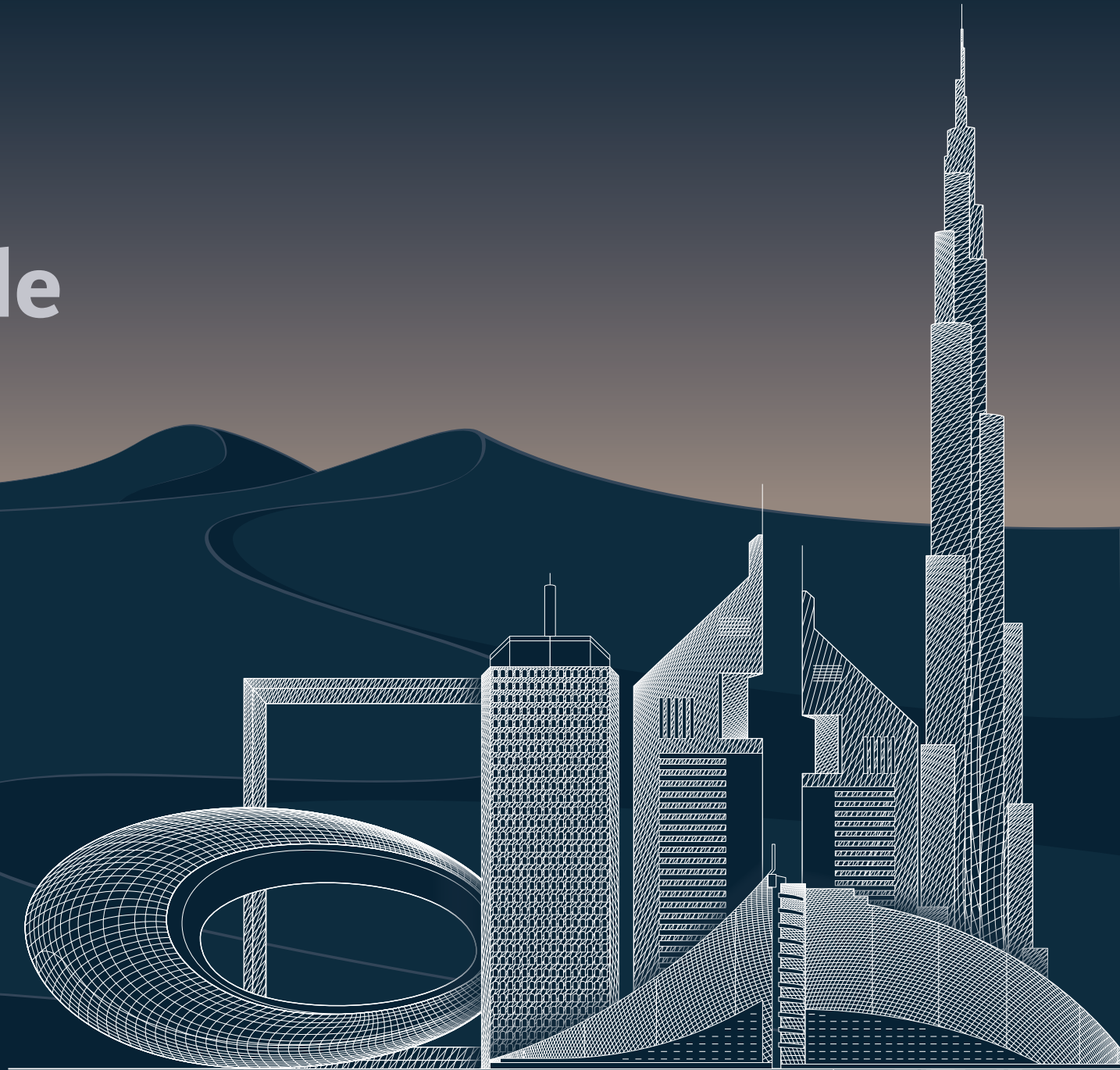


Dubai Building Code

2021 Edition



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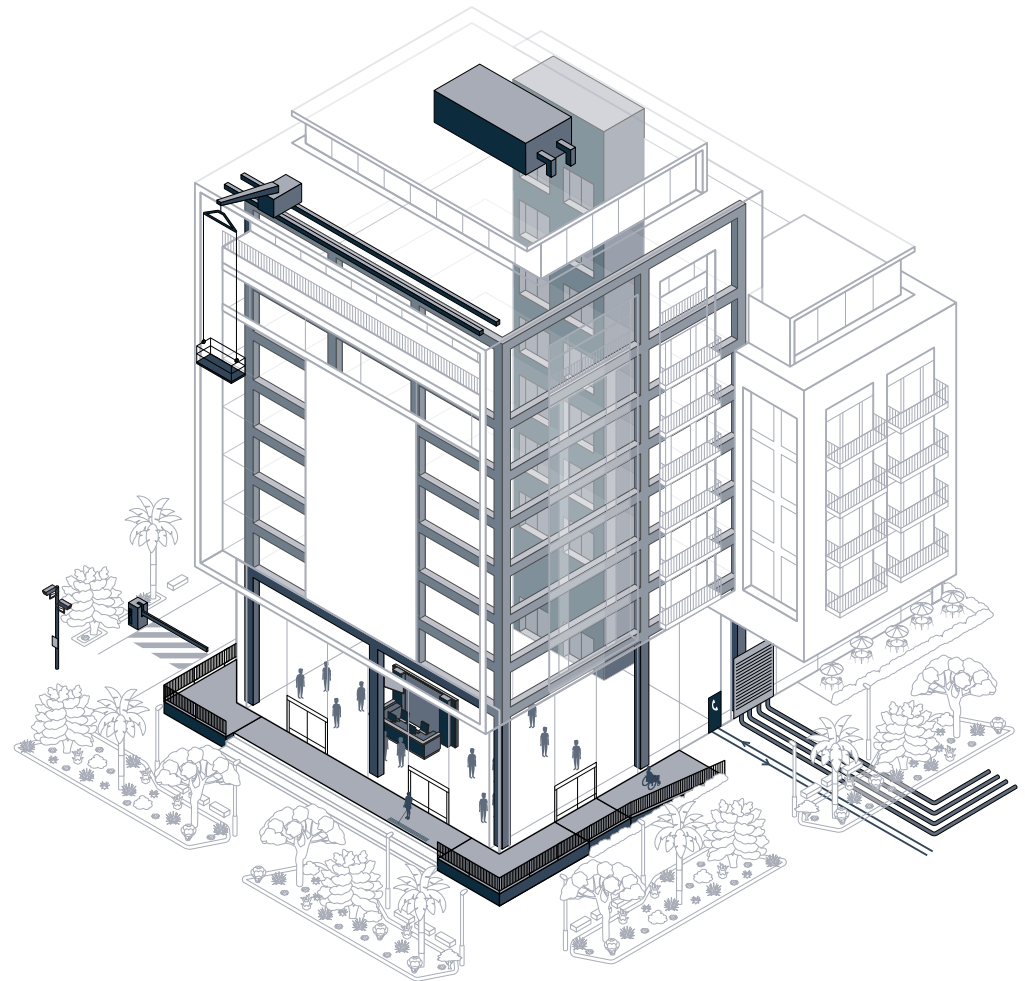
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Part A

General

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A.1 Introduction

The objective of the Dubai Building Code (DBC) is to unify building design across Dubai, and to create a building code that is easy to use and clearly mandates the minimum requirements for:

- a) the health, safety, welfare and convenience of people in and around buildings;
- b) the health, safety, welfare and convenience of people who might be affected by buildings;
- c) building design to reduce the impact on the surrounding environment; and
- d) the sustainable development of buildings.

The content of the DBC is based on the following inputs:

- 1) existing regulations and technical requirements produced by the various Authorities and Service Providers;
- 2) interviews with Government Authorities, Service Providers, Master Developers, Consultants and other stakeholders to understand gaps or inconsistencies between existing regulations;
- 3) benchmarking to international and regional codes.

The DBC is arranged under themes to integrate all relevant elements of building design.

The regulations stipulated in the DBC are the minimum requirements for building design. Nothing in the DBC prevents a building design that achieves higher standards than the minimum set out in the DBC.

A.2 Definitions

NOTE: This section gives common definitions that apply throughout the DBC. Each Part also has a section for terms specific to that Part.

A.2.1 Terms

Addition: Increase to a building's gross area or built-up area, and/or height or depth.

Architect: Physical or legal person in charge of designing or supervising building works, or both, and licensed to practise the architecture profession in Dubai in accordance with the applicable legislation.

Authority: Any ministry, department, entity, or federal or local government institution in Dubai or any department in licensing Authorities related to the implementation and the provisions of the DBC.

Building lifecycle: Series of stages through which a building passes during its lifetime, from initial design, through construction, commissioning, handover, and operation, to final demolition and recycling of demolished materials/systems.

Built-up area: Total constructed area in a building or structure measured from the external walls of the building, inclusive of balconies, terraces and other projections, as well as any other covered spaces such as covered parking structures, circulation corridors, loading/unloading bays, basements, service floors, indoor swimming pools and any other permanent structures on the plot.

Building: Structure for the use or occupancy of people or for shelter.

Change of use: Alteration to the use of a building to accommodate an alternative occupancy to that which it was originally designed.

Civil Engineer: Physical or legal person in charge of designing or supervising civil engineering works, or both, and licensed to practise the engineering profession in Dubai in accordance with the applicable legislation.

Consultant: Physical or legal person in charge of providing consulting advice during design or construction, or both and licensed to practise consulting in Dubai in accordance with the applicable legislation.

Contractor: Physical or legal person in charge of carrying out construction works and licensed to practise contracting activities in Dubai in accordance with the applicable legislation.

Developer: Person or entity who prepares land for building sites, constructs buildings, creates residential subdivisions or private enterprises, and rehabilitates existing buildings.

Development: General term for the construction, erection or placing of a building or structure; the making of an addition or alteration to a building or structure; a significant change in use or in intensity of use of any building or structure.

Development control regulations (DCR): Document that shows planning regulations (such as height, land use, area, setbacks, building lines) for the full development, and each plot in a subdivision plan, as issued by the relevant planning Authority or master developer.

Engineer: Physical or legal person in charge of designing or supervising construction works, or both, and licensed to practise the engineering profession in Dubai in accordance with the applicable legislation.

Existing building: Building that is completed, including commissioning and handover. The DBC applies to renovation, modification, reconstruction, addition to, or change of use of an existing building.

Floor area ratio (FAR): Ratio of gross floor area (GFA) to plot area.

Gross area (GA): Floor area within the inside perimeter of the exterior walls of a building. The measurement excludes shafts and courtyards, but includes corridors, stairways, ramps, closets, base of atria (or similar voids) and the thickness of interior walls, columns or other features.

Gross floor area (GFA): Measurement used by Planning Authorities in Dubai to define the floor area ratio (FAR), where FAR is the ratio of GFA to plot area. The building GFA is the sum of the GFA of all floors, measured to the exterior surface of the external wall thickness and from the centrelines of the common walls joining two spaces. The measurement excludes external wall features such as corniches, brackets, and façade cladding materials.

Habitable space: Space in a building for living, sleeping or eating involving occupancy for continuous period of time. Bathrooms, toilet rooms, closets, halls, storage or utility spaces and similar areas are not habitable spaces.

Heritage building: New or existing building in a heritage area. The heritage department of the Authority controls the elevation design and façade treatment of buildings in these areas.

Historical building: Existing building of historical importance where the original character of the building is to be retained.

Masterplan: Comprehensive plan to guide the long-term physical development of a specific area based on identified objectives, strategies, and timelines for implementation. Includes plans for land use, community facilities and utilities, transportation of goods and people, and energy use and conservation. Zoning ordinances and policies are developed for different areas or zoning districts based on the masterplan.

Modification: Fit-out of a space or unit, or change to a service system or building element.

Net area (NA): Actual occupied area, not including unoccupied accessory areas such as corridors, stairways, ramps, toilet rooms, mechanical rooms and closets.

Occupiable space: Room or enclosed space designed for human occupancy in which individuals congregate for amusement, labour, educational or similar purposes.

Owner: Any person, entity, firm or corporation having any legal or equitable interest in the property in whose name the plot or building is registered as Owner, including persons who have a long-term lease for a specified period; the Main or Sub-Developer in relation to unsold lands and buildings.

Performance-based design: Design that follows prescriptive design requirements, but an aspect(s) of which is based on an alternative solution that is equivalent to or better than the prescriptive design requirements.

Plot: Area of land that is:

- a) clearly defined (by landmarks, coordinates, distinct number, and the lengths of its sides and location);
- a) intended for construction under any approved plan or division project, or by any other method; and
- a) legally authorized to be used for construction or building as one unit.

Plot area: Total area of a plot between its boundary lines as measured on a horizontal plane.

Prescriptive design: Design that meets the requirements entirely.

Project: Construction of a permanent building, or any other civil work on a leased property including any modifications or installations in pre-built facilities.

Reconstruction: Modification that is not standalone. It affects adjacent egress routes or systems that are not part of the reconstructed area, such that a larger part of the building might need to be closed down or not occupied for a period of time.

Renovation: Change to linings/finishes or strengthening of structure. It does not involve any reconfiguration of internal spaces.

Townhouse: Multiple villas connected by one or more walls to a series of similar villas.

Villa: Separate building located on a separate plot allocated with all its floors for the dwelling of one single family and an independent car parking space in addition to independent external open space.

A.3 References

NOTE: This section gives references that apply throughout the DBC. Each Part also has a section for references specific to that Part.

Ref. A.1 UAE MINISTRY OF THE INTERIOR GENERAL COMMAND OF CIVIL DEFENCE, 2018. UAE Fire and Life Safety Code of Practice (UAE FLSC). United Arab Emirates: Ministry of the Interior General Command of Civil Defence.

Ref. A.2 GOVERNMENT OF DUBAI, 2017. Dubai universal design code. Dubai: Government of Dubai.

Ref. A.3 DUBAI HEALTH AUTHORITY, 2012. Regulations (various). Dubai: Dubai Health Authority. Available at:
www.dha.gov.ae/en/HealthRegulation/Pages/FacilityRelated.aspx.

Ref. A.4 DUBAI HEALTH AUTHORITY, 2019. DHA Health facility guidelines – Part A to Part F. Dubai: Dubai Health Authority. Available at:
www.dubaitourism.gov.ae/en/services.

Ref. A.5 DEPARTMENT OF TOURISM AND COMMERCE MARKETING, 2019. DTCM Hotel classification criteria for hotels, guest houses and hotel apartments. Dubai: DTCM. Available at: classification.dubaitourism.ae.

Ref. A.6 DUBAI CIVIL AVIATION AUTHORITY Application for a no-objection certificate to construct a building (above and below 300 m) in Aviation Easement areas. Dubai: DCAA. Available at:
www.dcaa.gov.ae/services/building-constructions/construct-building.

Ref. A.7 GREAT BRITAIN, 2015. The Construction (Design and Management) Regulations. London: The Stationery Office.

A.3.1 Acronyms and abbreviations

ASCE	American Society of Civil Engineers
BIM	building information modelling
BS EN	British Standard European Norm
Ch.	chapter
DTCM	Department of Tourism and Commerce Marketing
DBC	Dubai Building Code
DCAA	Dubai Civil Aviation Authority
DCR	Development Control Regulations
DDA	Dubai Development Authority
DEWA	Dubai Electricity and Water Authority
DHA	Dubai Health Authority
DM	Dubai Municipality
DLTM	Dubai Local Transverse Mercator
DSOA	Dubai Silicon Oasis Authority
Emicool	Emirates District Cooling LLC
Empower	Emirates Central Cooling Systems Corporation
FAR	floor area ratio
G	ground floor
GA	gross area
GFA	gross floor area
IACAD	Islamic Affairs and Charitable Activities Department
KHDA	Knowledge and Human Development Authority
MEP	mechanical, electrical and plumbing
NA	net area

RTA	Roads and Transport Authority
SI	International system of units
SIRA	Security Industry Regulatory Agency
TRA	Telecommunications Regulatory Authority
Trakhees	Department of Planning and Development – Ports Customs and Freezone Corporation
UAE FLSC	UAE Fire and Life Safety Code of Practice

A.3.2 Verbal forms used within the DBC

May	The associated text is permitted
Shall	The associated text is mandatory
Should	The associated text is recommended

A.4 Relationship of DBC to other local and international regulations

A.4.1 Local regulations

The DBC incorporates, unifies and replaces the regulations related to building design published by Dubai Municipality (DM), Dubai Development Authority (DDA), Department of Planning and Development – Ports Customs and Freezone Corporation (Trakhees) and Dubai Silicon Oasis Authority (DSOA).

Regulations related to building design by Dubai Electricity and Water Authority (DEWA) and Islamic Affairs and Charitable Activities Department (IACAD) are incorporated or cross-referenced by the DBC.

Regulations related to building design by the Security Industry Regulatory Agency (SIRA) are summarized and cross-referenced in **Part J**.

Federal regulations also remain in place. The DBC follows the Federal regulations of Telecommunications Regulatory Authority (TRA) and the Ministry of Interior Civil Defence. The TRA regulations are given in **G.11**. The UAE Fire and Life Safety Code (UAE FLSC) **[Ref. A.1]** is cross-referenced throughout the DBC.

Accessibility requirements of the Dubai Universal Design Code **[Ref. A.2]** applicable to building design are incorporated in **Part C** and replace these aspects of the Dubai Universal Design Code.

The Dubai Health Authority (DHA) regulations **[Ref. A.3 and Ref. A.4]**, along with the relevant Federal regulations, shall be followed for the design of all healthcare facilities.

The Department of Tourism and Commerce Marketing (DTCM) sets out the requirements to achieve star ratings for hotels, resorts, hotel apartments and guest houses **[Ref. A.5]**. Achieving the minimum requirements of the DBC does not guarantee a rating from DTCM. DTCM should be consulted at the earliest stage of design.

The Knowledge and Human Development Authority (KHDA) and the Ministry of Education both license education facilities, but they do not provide separate design requirements. Education facilities are approved by the main Authorities in accordance with the DBC.

Roads and Transport Authority (RTA) regulations in relation to the right of way are referenced by the DBC as appropriate.

Dubai Civil Aviation Authority (DCAA) regulates maximum building heights within the flight path.

Regulations related to permitting, procedures, construction and design of non-building elements such as infrastructure, transportation systems, and the public realm remain, and can be obtained from, the relevant Authority.

Planning development regulations published by Developers or Planning Authorities are not replaced by the DBC. The DBC contains planning regulations in **Part B**, but these shall be followed only where plot-specific planning development regulations are unavailable.

A.4.2 International codes and standards

The DBC relies on international codes and standards, mainly BS EN and American codes. These codes and standards shall be used in their latest edition except where the DBC lists an edition.

NOTE: As an example, Part F (Structures) mandates the 2016 edition of ASCE/SEI 7.

An edition is mandated in the DBC for one of two reasons.

- a) The DBC refers to sections, tables and figures within that version of the international code or standard and therefore the cross-references might not apply to later editions.
- b) The edition has been reviewed for its applicability to Dubai.

A.4.3 Hierarchy of codes and standards

The hierarchy of the adopted codes and standards is defined as follows.

- a) The DBC and UAE Fire and Life Safety Code of Practice (UAE FLSC) **[Ref. A.1]** together form the main regulations for building design in Dubai. The UAE FLSC is referenced throughout the DBC.
- b) The DBC follows the SIRA regulations for the design of security. These regulations are summarized and referenced in **Part J**.
- c) The DBC follows the Federal regulations of TRA for the design of telecommunications. These regulations are documented in **G.11**.
- d) The international codes and standards for design, material and product specification listed under the heading of “Essential references” in each Part, and referenced in the relevant sections of each Part, are also requirements of the DBC.
- e) Documents listed under the heading of “Further reading” in each Part are provided for information as useful guides to building design.

Where there is conflict between the DBC and reference model codes or standards, the most restrictive/highest performance requirements shall be met.

Where there is a conflict between a general requirement and a specific requirement of the DBC or its referenced codes/standards then the specific requirement applies.

A.4.4 Safe by design

The principle of “safe by design” requires the Consultant to address the health and safety of all occupants, Contractors and users of the building. This includes Contractors, maintenance staff, occupants, visitors and demolition teams throughout the building lifecycle.

This process is documented in a risk register where the Consultant acknowledges the risk, investigates mitigation measures and provides a design solution which reduces the risk to a practicable level, while also taking account of cost, constructability, other project goals, etc.

Safe by design principles are not mandated by the first edition of the DBC, but are likely to be enforced in future revisions. Consultants should familiarize themselves with these principles by reference to the Construction Design and Management Regulations **[Ref. A.7]**.

A.5 Scope and application of the DBC

A.5.1 General

The DBC applies to new buildings and changes to existing buildings as described in Table A.1, except where an alternative solution is permitted (see A.8).

Change to an existing building	Description	Applicable codes
Renovation	Change to linings/finishes or strengthening of structure. It does not involve any reconfiguration of internal spaces.	DBC and UAE FLSC [Ref. A.1] apply to interior finish changes or building envelope changes.
Modification	Fit-out of a space/unit, or change to a service system or building element.	DBC and UAE FLSC [Ref. A.1] apply to the fit-out or change.
Reconstruction	Modification that is not stand-alone. It affects adjacent egress routes or systems that are not part of the reconstruction area such that a larger part of the building might need to be closed down or not occupied for a period of time.	DBC and UAE FLSC [Ref. A.1] apply to the reconstruction and any areas of the building which are impacted by the reconstruction.
Change of use	Change of use or occupancy of part or all of the building.	DBC and UAE FLSC [Ref. A.1] apply to the change of use and any areas of the building which are impacted by the change of use, e.g. existing exit stairs and routes, vertical transport, quantity and type of sanitary provisions.
Addition	An increase in the building's gross area or built-up area, and/or height or depth.	DBC and UAE FLSC [Ref. A.1] apply to the addition. If the addition relies on the remainder of the existing building for escape, vertical transport, etc., the existing building shall also be assessed against the DBC and UAE FLSC [Ref. A.1].

Table A.1 Existing building and application of the DBC

The DBC applies to the occupancies and use types described in Table A.2.

Occupancy	Use
Assembly	Amusement park, theatre, cinema, restaurant, museum, art gallery, place of worship, library, exhibition and conference centre, etc.
Business	Office, professional services, government centre, post office, bank, etc.
Educational	University, college, school, kindergarten, nursery, educational institute
Hotel establishment	Hotel, resort, guesthouse, hotel apartment
Residential	Apartment, studio, student accommodation, labour accommodation, staff accommodation Villas and townhouses
Parking	Open, enclosed, mechanical
Retail	Shopping centre, department store, shop, kiosk, showroom, stores
Mall	Open mall, enclosed mall
Industrial	Factory, workshop
Storage	Warehouse

Table A.2 Occupancies and use types

The DBC does not apply to:

- a) infrastructure such as road and rail bridges, tunnels, culverts, metro stations, hydraulic structures, buried and overhead utilities, power stations, antennas and masts, wind turbines and nuclear facilities;
- b) marine structures such as canals, dams, dikes, levies, piers, jetties and wharves;
- c) oil and gas structures such as petrochemical facilities, oil refineries, liquefied natural gas terminals, oil platforms and tanks;
- d) structures such as tents, cranes, storage racks, scaffolding, formwork and falsework;
- e) structures subject to specific loading conditions such as stadia, grandstands, silos and chimneys.

The DBC does not apply to historical buildings where the original character of the building is to be retained. Buildings considered to be historical shall conform to the guidelines of the relevant Authority as appropriate.

Buildings in heritage areas or considered to be heritage buildings shall follow the DBC. The heritage department of the Authority controls the elevation design and façade treatment.

Villas and townhouses are addressed in **Part K**.

A.5.2 Floor area and building height definitions in DBC

A.5.2.1 Gross floor area (GFA)

Gross floor area (GFA) is used by Planning Authorities in Dubai to define the floor area ratio (FAR), where FAR is the ratio of GFA to plot area. The building GFA is the sum of the GFA of all floors, measured to the exterior surface of the external wall thickness and from the centrelines of the common walls joining two spaces. The measurement excludes external wall features such as corniches, brackets, and façade cladding materials.

GFA is calculated based on Table A.3.

GFA might be in development control regulations (DCR). Consultants shall calculate GFA using the DBC definition and Table A.1 unless GFA is stated in the DCR.

Space or area	Included or excluded in GFA
Habitable and occupiable spaces including sanitary facilities	Included
Mezzanine floors	Included
Lobbies	Included
Corridors	Included
Stairs	Included
Basements in buildings other than villas	Habitable and occupiable spaces including sanitary facilities to be included
Basements in villas	Excluded
Attic spaces	Attic spaces less than 2.15 m height and used for services only to be excluded
Parking	Excluded
Parking lobbies	Excluded
Internal roads	Excluded
Loading areas	Excluded
Mechanical and service rooms	Excluded
Mechanical ducts	Excluded
Elevator shaft at elevator recall floor	Included
Elevator shafts at other floors	Excluded
Elevator machine rooms	Excluded
Mechanical, electrical and plumbing plant floors	Excluded
Shafts	Excluded
Refuse chutes	Excluded
Waste rooms	Excluded
Storage below ground	Excluded

Table A.3 Areas included in GFA

Space or area	Included or excluded in GFA
Storage on ground and above	Included
Prayer room and ablution area	Excluded
Gym with gross area $\leq 50\%$ of roof area	Excluded
Gym with gross area $> 50\%$ of roof area	The gross area of the gym exceeding 50% is included*
Swimming pools (indoor and outdoor) including services (showers, toilets, lockers, changing rooms, etc.)	Excluded
Arcades (open, non-air conditioned)	Excluded
Base of double height volumes and atria	Included
Void of double height floors and atria	Excluded
Floor voids	Excluded
Terraces	Excluded
Balconies	Excluded
NOTE: * This rule is associated with the roof area but it applies to gyms located anywhere in the building.	

A.5.2.2 Built-up area

Built-up area is used also by Planning Authorities. It is defined as the total constructed area in a building or structure measured from the external walls of the building, inclusive of balconies, terraces and other projections as well as any other covered spaces such as covered parking structures, circulation corridors, loading/unloading bays, basements, service floors, indoor swimming pools and any other permanent structures on the plot. Calculation of total built-up area shall exclude the following:

- a) shafts, courtyards and yards including mechanical, electrical and plumbing (MEP) shafts, refuse chutes and similar;
- b) non-covered outdoor landscape and hardscape surfaces and courtyards;
- c) lightweight outdoor shading structures such as pergolas and standalone canopies.

An example calculation using a sample plan is shown in Figure A.1.

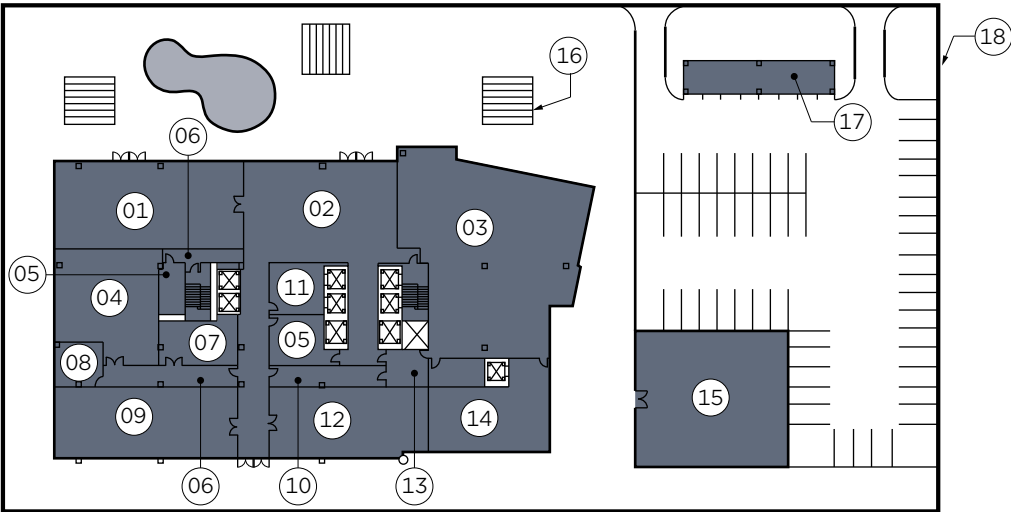


Figure A.1 Total built-up area (highlighted areas are included in area definition)

Key		
01: Gym	07: Staff toilets	13: Back-of-house
02: Lobby	08: Staff	14: Kitchen
03: Ballroom	09: Meeting room	15: Service block
04: Prayer room	10: Service corridor	16: Pergolas
05: MEP	11: Electrical	17: Parking canopy
06: Corridor	12: Restaurant	18: Plot limit

A.5.2.3 Gross area and net area

The terms “gross area” (GA) and “net area” (NA) are used by the DBC to calculate occupant load in a floor or space. They are defined in A.2 and further described and illustrated in [Part B](#).

A.5.2.4 Building height

Building height is a key variable that is checked by the Authorities for compliance with the DBC, DCR/affection plans, UAE FLSC [\[Ref. A.1\]](#) and, where applicable, DCAA restrictions on building heights located in Aviation easement areas due to overhead flight paths.

Each definition of building height is slightly different as they are used to regulate different issues. For this reason, Consultants shall check building height against the DBC, DCR/affection plans, UAE FLSC [\[Ref. A.1\]](#) and, where applicable, DCAA regulations [\[Ref. A.6\]](#).

A.6 Navigating the DBC

A.6.1 Division into Parts

The DBC is divided into Parts as described in Figure A.2. Each Part is subdivided into sections by topic.

Minimum requirements for sustainability are embedded within the relevant Parts of the DBC. Developers and Consultants are encouraged to apply the rating system adopted by the relevant Authority to assess their projects and exceed these minimum requirements.

Fire safety requirements are given in the relevant Parts of the DBC and cross referenced to UAE FLSC [Ref. A.1]. The aim of this cross-referencing is to help Consultants navigate and coordinate the requirements of the two documents. Where an element of UAE FLSC is not applicable in Dubai, or the DBC contains additional information not in UAE FLSC, this is indicated in the text.

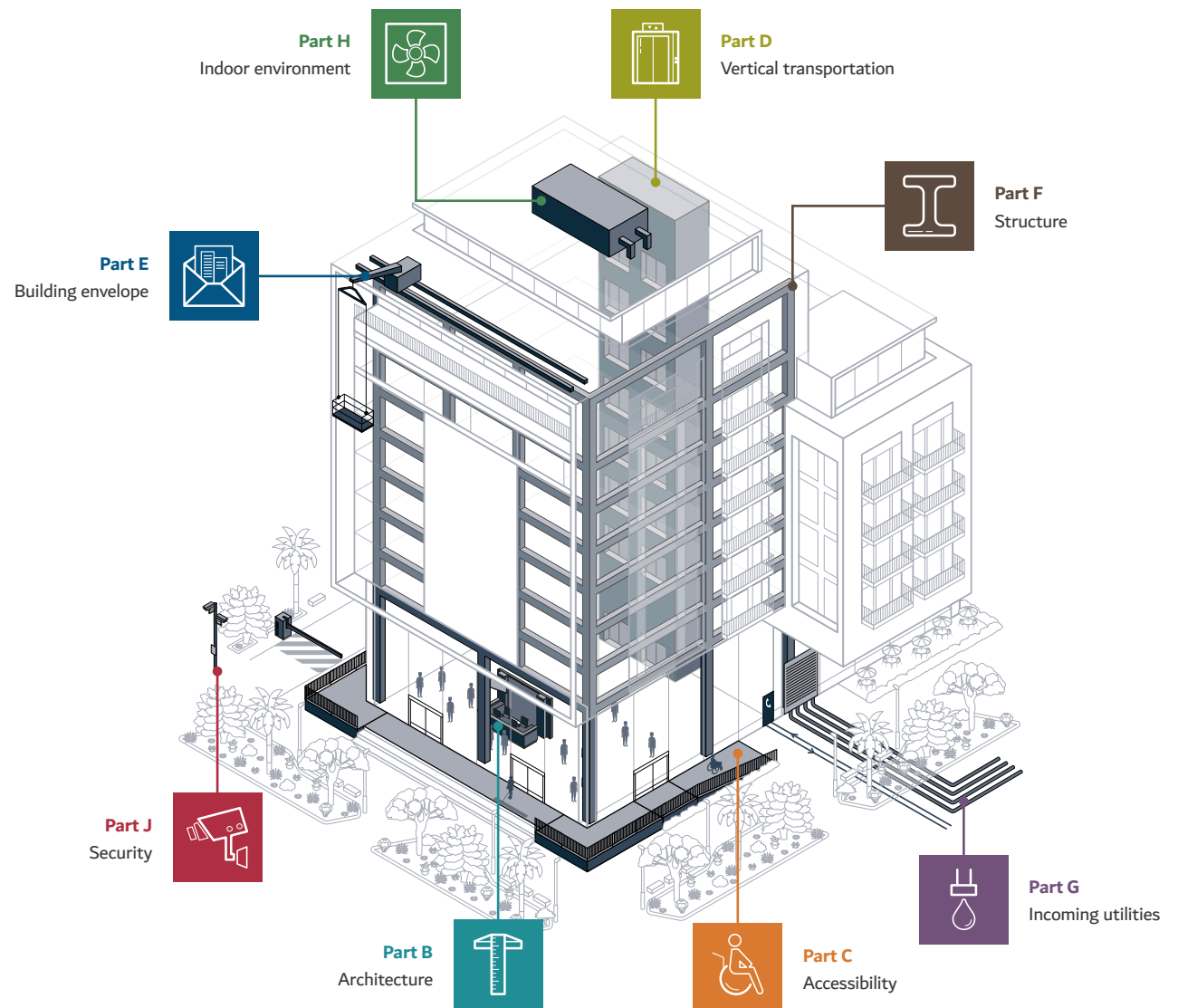


Figure A.2 How to use and navigate the DBC

A.6.2 Performance statements

The overarching objectives of the DBC, its intention and regulatory scope, are set out in A.1.

Each Part of the DBC starts with a set of performance statements. These set out the broad outcomes that the completed building is expected to achieve, and are provided to explain the intent of the prescriptive design requirements within the Part. If the prescriptive design requirements of the Part are followed, then the performance statements of the Part are deemed to be met and will be accepted by the Authority.

A.6.3 References

Each Part has a section for essential references and further reading. The requirements of the DBC and the essential references form the prescriptive design requirements of the DBC and will be enforced by the Authority. Websites and webpages shall be checked regularly for updates.

NOTE: Where websites and webpages have been cited throughout the DBC, they are provided for ease of reference and are correct at the time of publication. Changes to the location of a webpage or website, or its contents, cannot be guaranteed.

A.6.4 Definitions

In addition to the common definitions in A.2, each Part has a section for definitions.

A.6.5 Annexes

Any annex material is included at the end of the Part to which it relates. Annex material contains further explanatory material.

A.7 Adoption of DBC

Projects in design or construction that have a final approved design from the Authority do not need to conform to the DBC.

Projects in design without a final approved design shall conform to the DBC. The requirements of the DBC are based on regulations enforced prior to its publication and in many instances the DBC is less stringent. For example, requirements unrelated to health, safety, welfare and environment have generally been removed from the regulations. Therefore, it is expected that there will be benefits to Developers and Owners in adopting the DBC.

A.8 Alternative solutions

The DBC is intended to be a prescriptive design code. Performance statements are provided at the start of each Part to describe the outcomes that the completed building is expected to achieve.

The performance statements form a set of overarching regulations that convey the requirements of the DBC (see A.6.2).

The preferred method of meeting the performance statements of the DBC is to follow the prescriptive design requirements in each Part.

The inclusion of performance statements sets out the intent of the DBC and future-proofs it, supporting a possible transition to a more performance-based regulatory system in the future.

Alternative solutions which meet the performance statements of the DBC are permitted:

- a) for projects that cannot reasonably meet the prescribed requirements; and
- b) for changes to existing buildings where the requirements of the DBC would result in a disproportionate cost to the Owner/Developer.

The Authority will decide if an alternative solution is permitted.

An alternative solution shall meet the intent of the performance statements and provide at least an equivalent level of performance to all of the prescriptive design requirements.

Alternative solutions shall be applied to a performance-based design holistically such that a change from one Part of the DBC does not reduce the building's performance in other aspects.

Alternative solutions for changes to existing buildings shall demonstrate that the existing conditions are maintained to the current level of compliance or improved to meet the safety levels of the DBC and UAE FLSC **[Ref. A.1]**.

The Authority will decide whether or not a performance-based design is equivalent to the prescriptive design requirements. The responsibility is on the Consultant to demonstrate that the performance-based design is required and that it is at least equivalent to the prescriptive design requirements.

The DBC lists preferred international codes and standards in each Part (see A.4.2). Where permitted by the Authority, alternative standards can be used provided that the comparable standard meets the intent of the DBC.

A.9 Maintenance of DBC

A.9.1 General

The DBC is an interactive document available electronically.

The DBC will typically be maintained and updated on a two- to three-year cycle. If necessary, addendums will be issued between cycles. It is therefore important that Consultants, Contractors and Engineers check back regularly to ensure that they are following the latest version of the DBC.

A.9.2 Latest changes to DBC

This edition is the first edition of the DBC.

Amended content in future editions will be clearly indicated. For example, new content will be side lined in the first edition in which it is published.

A.10 Copyright

No content of the DBC, in part or whole, shall be copied, printed, sold or reproduced in any format.

A.11 Coordinate system

The coordinate system used by the Authority is Dubai Local Transverse Mercator (DLTM). The reference point for levelling is Port Rashid.

A.12 Dimensions, units and calculations

A.12.1 Dimensions

All dimensions in the figures and diagrams of the DBC are given in millimetres without units, unless the distance exceeds 1,000 mm, and then it is given in metres with units (m). Dimensions that are not stated as “maximum” or “minimum” are absolute.

All dimensions are subject to conventional industry tolerances, except where the requirement is stated as a range with specific minimum and maximum end points.

A.12.2 Units

All quantities shall be in metric units with the units defined.

All engineering shall be performed in dynamically consistent units. One of the two systems shown in Table A.4 should be used. Use of dimensions between analysis models, drawings and/or building information modelling (BIM) models shall be fully coordinated and consistent.

A.12.3 Calculations

Where the required number of elements (e.g. parking or sanitary provisions) to be provided is determined by calculations of ratios or percentages, and this results in remainders or fractions, the next greater whole number of such elements shall be provided.

Unit	SI-system symbol (preferred units)	System mm-t-s		System mm-kg-ms	
		Symbol	Conversion from SI units	Symbol	Conversion from SI units
Length	m	mm	10^3	mm	10^3
Mass	kg	t	10^{-3}	kg	1
Time	s	s	1	ms	10^3
Temperature	°C	°C	1	°C	1
Energy	J	mJ	10^3	J	1
Acceleration	m/s^2	mm/s^2	10^3	mm/ms^2	10^{-3}
Area	m^2	mm^2	10^6	mm^2	10^6
Frequency	Hz	Hz	1	ms^{-1}	10^{-3}
Velocity	m/s	mm/s	10^3	mm/ms	1
Volume	m^3	mm^3	10^9	mm^3	10^9
Density	kg/m^3	t/mm^3	10^{-12}	kg/mm^3	10^{-9}
Stress	N/m^2	N/mm^2	10^{-6}	kN/mm^2	10^{-9}
Force	N	N	1	kN	10^{-3}
Moment	Nm	Nmm	10^3	kNmm	1
Stiffness	N/m	N/mm	10^{-3}	kN/mm	10^{-6}

Table A.4 Preferred units